

Compare Results

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New File:

USB_PD_R3_2 V1.1 2024-10_Chapter 5 PHY Layer.pdf

35 pages (1.38 MB)

28/10/2024 11:27:35

Total Changes

666

Text only comparison

Content

524 Replacements

67 Insertions

75 Deletions

Styling and Annotations

0 Styling

0 Annotations

Go to First Change (page 1)

5 Physical Layer

5.1 Physical Layer Overview

The Physical Layer (*PHY Layer*) defines the *Signaling* technology for USB Power Delivery. This chapter defines the electrical requirements and parameters of the *PHY Layer* required for interoperability between *PDUSB Devices*.

5.2 Physical Layer Functions

The USB PD *PHY Layer* consists of a pair of transmitters and receivers that communicate across a single signal wire (*CC*). All communication is half duplex. The *PHY Layer* practices *Collision Avoidance* to minimize communication errors on the channel.

The transmitter performs the following functions:

- Receive *Packet* data from the *Protocol Layer*.
- Calculate and append a *CRC*.
- Encode the *Packet* data including the *CRC* (i.e., the *Payload*).
- Transmit the *Packet* (*Preamble*, *SOP**, *Payload*, *CRC* and *EOP*) across the channel using *Bi-phase Mark Coding* (*BMC*) over *CC*.

The receiver performs the following functions:

- Recover the clock and lock onto the *Packet* from the *Preamble*.
- Detect the *SOP**.
- Decode the received data including the *CRC*.
- Detect the *EOP* and validate the *CRC*:
 - If the *CRC* is **Valid**, deliver the *Packet* data to the *Protocol Layer*.
 - If the *CRC* is **Invalid**, flush the received data.

5.3 Symbol Encoding

Except for the *Preamble*, all communications on the line **shall** be encoded with a line code to ensure a reasonable level of DC-balance and a suitable number of transitions. This encoding makes receiver design less complicated and allows for more variations in the receiver design.

4b5b line code **shall** be used. This encodes 4-bit data to 5-bit symbols for transmission and decodes 5-bit symbols to 4-bit data for consumption by the receiver.

The 4b5b code provides data encoding along with special symbols. Special symbols are used to signal *Hard Reset*, and delineate *Packet* boundaries (see [Table 5.1, "4b5b Symbol Encoding"](#)).

Table 5.1 4b5b Symbol Encoding

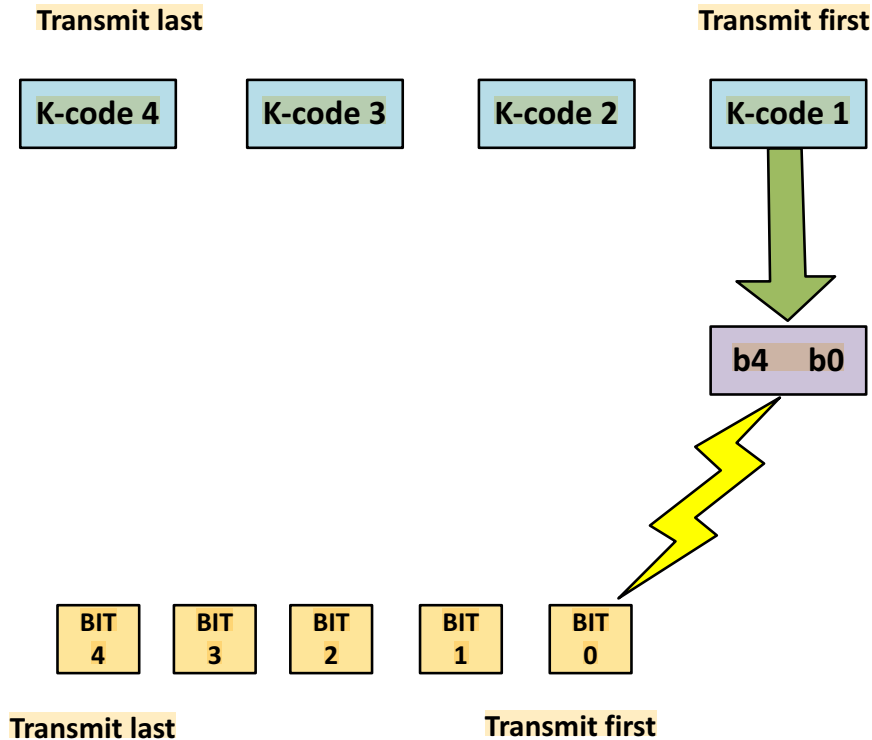
Name	4b	5b Symbol	Description
0	0000	11110	hex data 0
1	0001	01001	hex data 1
2	0010	10100	hex data 2
3	0011	10101	hex data 3
4	0100	01010	hex data 4
5	0101	01011	hex data 5
6	0110	01110	hex data 6
7	0111	01111	hex data 7
8	1000	10010	hex data 8
9	1001	10011	hex data 9
A	1010	10110	hex data A
B	1011	10111	hex data B
C	1100	11010	hex data C
D	1101	11011	hex data D
E	1110	11100	hex data E
F	1111	11101	hex data F
Sync-1	<i>K-code</i>	11000	Startsynch #1
Sync-2	<i>K-code</i>	10001	Startsynch #2
RST-1	<i>K-code</i>	00111	<i>Hard Reset</i> #1
RST-2	<i>K-code</i>	11001	<i>Hard Reset</i> #2
EOP	<i>K-code</i>	01101	<i>EOP</i> End of Packet
	Error	00000	Shall Not be used
	Error	00001	Shall Not be used
	Error	00010	Shall Not be used
	Error	00011	Shall Not be used
	Error	00100	Shall Not be used
	Error	00101	Shall Not be used
Sync-3	<i>K-code</i>	00110	Startsynch #3
	Error	01000	Shall Not be used
	Error	01100	Shall Not be used
	Error	10000	Shall Not be used
	Error	11111	Shall Not be used

5.4 Ordered Sets

Ordered sets **shall** be interpreted according to [Figure 5.1, "Interpretation of ordered sets"](#).

An ordered set consists of 4 *K-codes* sent as shown in [Figure 5.1, "Interpretation of ordered sets"](#).

Figure 5.1 Interpretation of ordered sets



A list of the ordered sets used by USB Power Delivery can be seen in [Table 5.2, "Ordered Sets"](#). *SOP** is a generic term used in place of *SOP/SOP'/SOP''*.

Table 5.2 Ordered Sets

Ordered Set	Reference
Cable Reset	Section 5.6.5, "Cable Reset"
Hard Reset	Section 5.6.4, "Hard Reset"
SOP	Section 5.6.1.2.1, "Start of Packet Sequence (SOP)"
SOP'	Section 5.6.1.2.2, "Start of Packet Sequence Prime (SOP')"
SOP'_Debug	Section 5.6.1.2.4, "Start of Packet Sequence Prime Debug (SOP'_Debug)"
SOP''	Section 5.6.1.2.3, "Start of Packet Sequence Double Prime (SOP'')"
SOP''_Debug	Section 5.6.1.2.5, "Start of Packet Sequence Double Prime Debug (SOP''_Debug)"

The receiver **shall** search for all four *K-codes*. When the receiver finds all four *K-codes* in the correct place, it **shall** interpret this as a **valid** ordered set. When the receiver finds three out of four *K-codes* in the correct place, it **may**

interpret this as a **Valid** ordered set. The receiver **Should** ensure that all four *K-codes* are **Valid** to avoid ambiguity in detection (see [Table 5.3, "Validation of Ordered Sets"](#)).

Table 5.3 Validation of Ordered Sets

	1st code	2nd code	3rd code	4th code
Valid ¹	Corrupt	<i>K-code</i>	<i>K-code</i>	<i>K-code</i>
Valid ¹	<i>K-code</i>	Corrupt	<i>K-code</i>	<i>K-code</i>
Valid ¹	<i>K-code</i>	<i>K-code</i>	Corrupt	<i>K-code</i>
Valid ¹	<i>K-code</i>	<i>K-code</i>	<i>K-code</i>	Corrupt
Valid ² (perfect)	<i>K-code</i>	<i>K-code</i>	<i>K-code</i>	<i>K-code</i>
Invalid (example)	<i>K-code</i>	Corrupt	<i>K-code</i>	Corrupt
1)	May be interpreted as a Valid ordered set.			
2)	Shall be interpreted as a Valid ordered set.			

5.5 Transmitted Bit Ordering

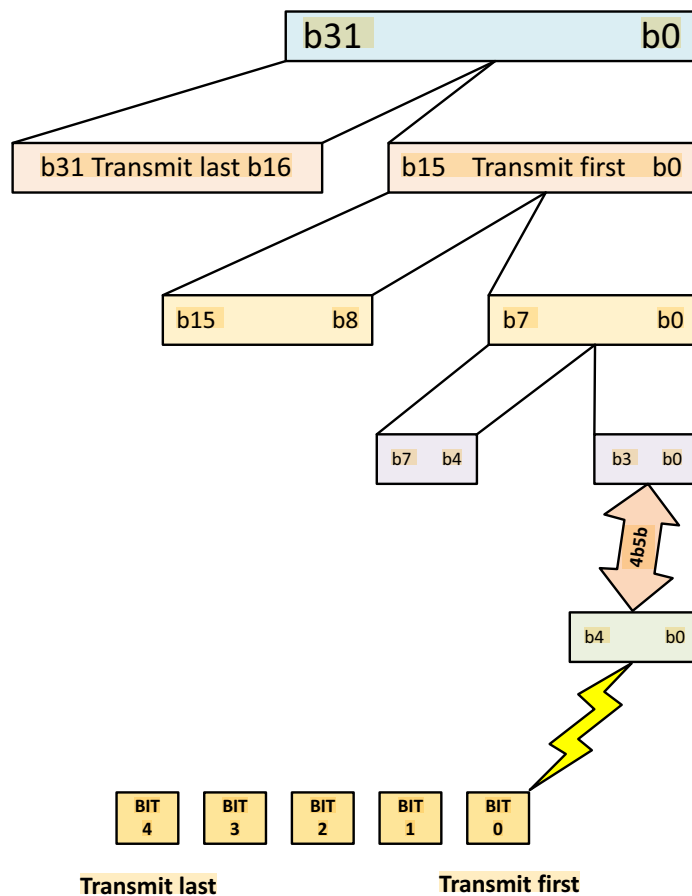
This section describes the order of bits on the wire that **shall** be used when transmitting data of varying sizes. [Table 5.4, "Data Size"](#) shows the different data sizes that are possible.

[Figure 5.2, "Transmit Order for Various Sizes of Data"](#) shows the transmission order that **shall** be followed.

Table 5.4 Data Size

	Unencoded	Encoded
Byte	8-bits	10-bits
Word	16-bits	20- bits
DWord	32-bits	40-bits

Figure 5.2 Transmit Order for Various Sizes of Data



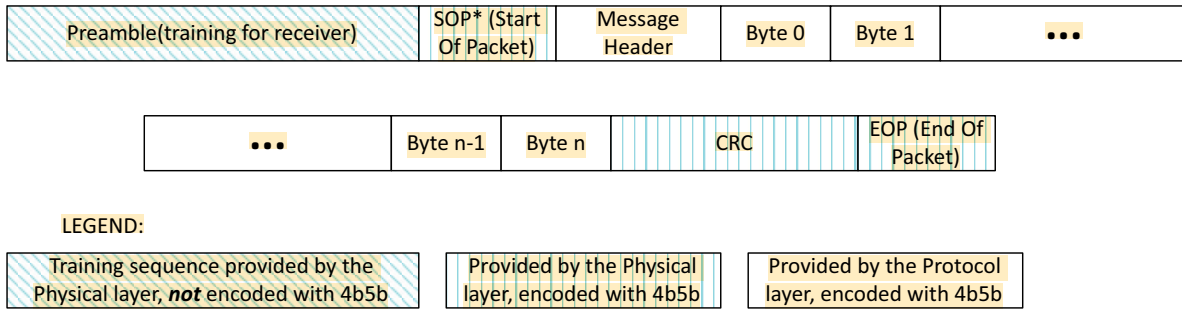
5.6 Packet Format

The *Packet* format **shall** consist of a *Preamble*, an *SOP**, (see [Section 5.6.1.2, "Start of Packet Sequences"](#)), *Packet* data including the *Message Header*, a *CRC* and an *EOP* (see [Section 5.6.1.5, "End of Packet \(EOP\)"](#)). The *Packet* format is shown in [Figure 5.3, "USB Power Delivery Packet Format"](#) and indicates which parts of the *Packet* **shall** be 4b/5b encoded. Once 4b/5b encoded, the entire *Packet* **shall** be transmitted using *BMC* over *CC*.

Note: All the bits in the *Packet*, including the *Preamble*, are *BMC* encoded.

See [Section 6.2.1, "Message Construction"](#) for more details of the *Packet* construction for *Control Messages*, *Data Messages* and *Extended Messages*.

Figure 5.3 USB Power Delivery Packet Format



5.6.1 Packet Framing

The transmission starts with a *Preamble* that is used to allow the receiver to lock onto the carrier. It is followed by a *SOP** (*Start of Packet*). The *Packet* is terminated with an *EOP* (*End of Packet*) *K-code*.

5.6.1.1 Preamble

The *Preamble* is used to achieve lock in the receiver by presenting an alternating series of "0s" and "1s", so the average frequency is the carrier frequency. Unlike the rest of the *Packet*, the *Preamble* **shall** Not be 4b/5b encoded.

The *Preamble* **shall** consist of a 64-bit sequence of alternating 0s and 1s. The *Preamble* **shall** start with a "0" and **shall** end with a "1".

5.6.1.2 Start of Packet Sequences

5.6.1.2.1 Start of Packet Sequence (SOP)

SOP is an ordered set. The *SOP* ordered set is defined as: three *Sync-1* *K-codes* followed by one *Sync-2* *K-code* (see [Table 5.5, "SOP Ordered Set"](#)).

Table 5.5 SOP Ordered Set

K-Code Number	K-Code in Code Table
1	<i>Sync-1</i>
2	<i>Sync-1</i>
3	<i>Sync-1</i>
4	<i>Sync-2</i>

A Power Delivery Capable *Source* or *Sink* **shall** be able to detect and communicate with *Packets* using *SOP*. If a *Valid* *SOP* is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **shall** be **Discarded**.

Sending and receiving of *SOP* *Packets* **shall** be limited to *PD Capable Ports* on *PDUSB Hosts* and *PDUSB Devices*. *Cable Plugs* and *VPDs* **shall** neither send nor receive *SOP* *Packets*.

Note: PDUSB Devices, even if they have the physical form of a cable (e.g., AMAs), are still required to respond to SOP Packets.

5.6.1.2.2 Start of Packet Sequence Prime (SOP')

The **SOP'** ordered set is defined as: two **Sync-1** K-codes followed by two **Sync-3** K-codes (see [Table 5.6, "SOP' Ordered Set"](#)).

Table 5.6 SOP' Ordered Set

K-Code Number	K-Code in Code Table
1	Sync-1
2	Sync-1
3	Sync-3
4	Sync-3

A VPD **Shall** have SOP' Communication capability. A VPD and a Cable Plug capable of SOP' Communications **Shall** only detect and communicate with **Packets** starting with **SOP'**.

A Port needing to communicate with a Cable Plug capable of SOP' Communications, Attached between a Port Pair will be able to communicate using both **Packets** starting with **SOP'** to communicate with the Cable Plug and starting with **SOP** to communicate with its Port Partner.

For a VPD or a Cable Plug supporting SOP' Communications, if a **Valid SOP'** is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **Shall** be **Discarded**. For a Port supporting SOP' Communications if a **Valid SOP** or **SOP'** is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **Shall** be **Discarded**. When there is no Explicit Contract or an Implicit Contract in place a Sink **Shall Not** send SOP' Packets and **Shall Discard** all **Packets** starting with **SOP'**.

5.6.1.2.3 Start of Packet Sequence Double Prime (SOP'')

The **SOP''** ordered set is defined as the following sequence of K-codes: **Sync-1, Sync-3, Sync-1, Sync-3** (see [Table 5.7, "SOP'' Ordered Set"](#)).

Table 5.7 SOP'' Ordered Set

K-Code Number	K-Code in Code Table
1	Sync-1
2	Sync-3
3	Sync-1
4	Sync-3

A VPD **Shall Not** have SOP'' Communication capability. A Cable Plug capable of SOP'' Communication, **Shall** have a SOP' Communication capability in the other Cable Plug. No cable **Shall** only support SOP'' Communication. A Cable Plug to which SOP'' Communication is assigned **Shall** only detect and communicate with **Packets** starting with **SOP''** and **Shall Discard** any other **Packets**.

A Port needing to communicate with such a Cable Plug, Attached between a Port Pair will be able to communicate using **Packets** starting with **SOP'** and **SOP''** to communicate with the Cable Plugs and **Packets** starting with **SOP** to communicate with its Port Partner. A Port which supports SOP'' Communication **Shall** also support SOP' Communication and **Shall** co-ordinate SOP* Communication so as to avoid collisions.

For the Cable Plug supporting SOP'' Communication, if a **Valid SOP''** is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **Shall** be **Discarded**. For the Port if a **Valid SOP*** is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **Shall** be **Discarded**.

5.6.1.2.4 Start of Packet Sequence Prime Debug (SOP' Debug)

The **SOP' Debug** ordered set is defined as the following sequence of *K-codes*: **Sync-1**, **RST-2**, **RST-2**, **Sync-3** (see [Table 5.8, "SOP' Debug Ordered Set"](#)). The usage of this Ordered Set is presently undefined.

Table 5.8 SOP' Debug Ordered Set

K-Code Number	K-Code in Code Table
1	Sync-1
2	RST-2
3	RST-2
4	Sync-3

5.6.1.2.5 Start of Packet Sequence Double Prime Debug (SOP" Debug)

The **SOP" Debug** ordered set is defined as the following sequence of *K-codes*: **Sync-1**, **RST-2**, **Sync-3**, **Sync-2** (see [Table 5.9, "SOP" Debug Ordered Set"](#)). The usage of this Ordered Set is presently undefined.

Table 5.9 SOP" Debug Ordered Set

K-Code Number	K-Code in Code Table
1	Sync-1
2	RST-2
3	Sync-3
4	Sync-2

5.6.1.3 Packet Payload

The **Packet** data is delivered from the **Protocol Layer** (see [Section 6.2, "Messages"](#)) and **Shall** be encoded with the hex data codes from [Table 5.1, "4b5b Symbol Encoding"](#).

5.6.1.4 CRC

The **CRC** **Shall** be inserted just after the **Payload**. It is described in [Section 5.6.2, "CRC"](#).

5.6.1.5 End of Packet (EOP)

The end of **Packet** marker **Shall** be a single **EOP** *K-code* as defined in [Figure 5.1, "Interpretation of ordered sets"](#). This **Shall** mark the end of the **CRC**. After the **EOP**, the **CRC**-residual **Shall** be checked. If the **CRC** is not good, the whole transmission **Shall** be **Discarded**, if it is good, the **Packet** **Shall** be delivered to the **Protocol Layer**.

Note: An **EOP** **May** be used to prematurely terminate a **Packet** e.g., before sending **Hard Reset** Signaling.

5.6.2 CRC

The **Message Header** and data **Shall** be protected by a 32-bit **CRC**.

- **CRC**-32 protects the data integrity of the data **Payload**. **CRC**-32 is defined as follows:
- The **CRC**-32 polynomial **Shall** be = 04C1_1DB7h.
- The **CRC**-32 Initial value **Shall** be = FFFF_FFFFh.
- **CRC**-32 **Shall** be calculated for all bytes of the **Payload** not inclusive of any **Packet** framing symbols (i.e., excludes the **Preamble**, **SOP***, **EOP**).
- **CRC**-32 calculation **Shall** begin at byte 0, bit 0 and continue to bit 7 of each of the bytes of the **Packet**.
- The remainder of **CRC**-32 **Shall** be complemented.
- The residual of **CRC**-32 **Shall** be C704 DD7Bh.

This inversion of the *CRC*-32 remainder adds an offset of FFFF_FFFFh that will create a constant *CRC*-32 residual of C704_DD7Bh at the receiver side.

Note: The *CRC* implementation is identical to the one used in [\[USB 3.2\]](#).

[Figure 5.4, "CRC-32 Generation"](#) is an illustration of *CRC*-32 generation. The output bit ordering **shall** be as detailed in [Table 5.10, "CRC-32 Mapping"](#).

Figure 5.4 CRC-32 Generation

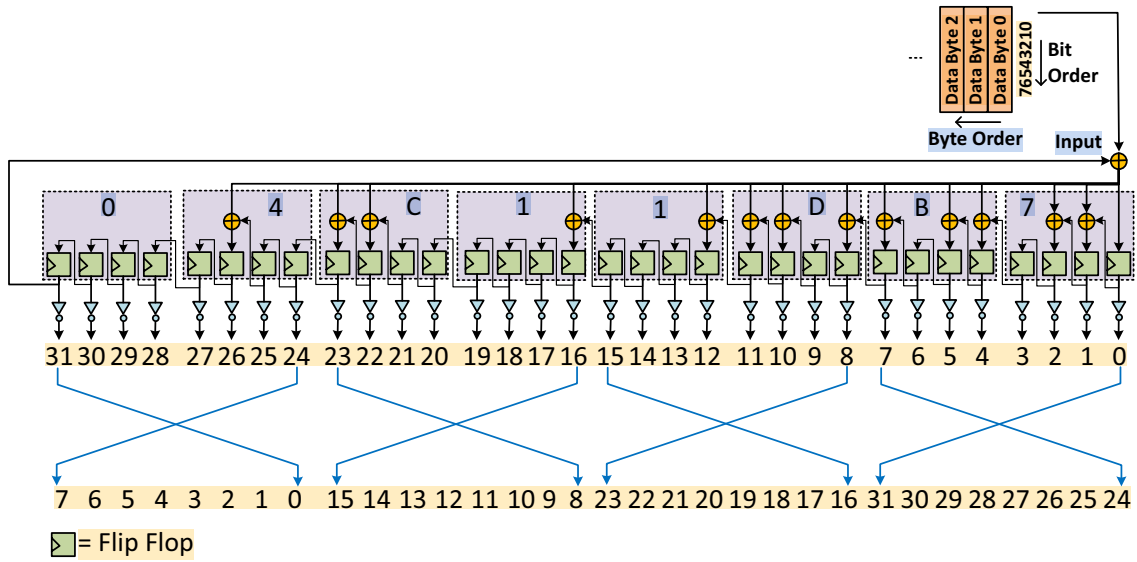


Table 5.10 CRC-32 Mapping

CRC-32	Result Bit Position in CRC-32 Field
0	31
1	30
2	29
3	28
4	27
5	26
6	25
7	24
8	23
9	22
10	21
11	20
12	19
13	18
14	17
15	16
16	15
17	14
18	13
19	12
20	11
21	10
22	9
23	8
24	7
25	6
26	5
27	4
28	3
29	2
30	1
31	0

The CRC-32 **Shall** be encoded before transmission.

5.6.3 Packet Detection Errors

CRC errors, or errors detected while decoding encoded symbols using the code table, **Shall** be treated the same way; the *Message* **Shall** be **Discarded** and a **GoodCRC** Message **Shall Not** be returned.

While the receiver is processing a *Packet*, if at any time the CC-line becomes *Idle* the receiver **Shall** stop processing the *Packet* and **Discard** it (no **GoodCRC** Message is returned). See [Section 5.8.6.1, "Definition of Idle"](#) for the definition of *BMC Idle*.

5.6.4 Hard Reset

Hard Reset Signaling is an ordered set of bytes sent with the purpose to be recognized by the *PHY Layer*. The **Hard Reset Signaling** ordered set is defined as: three **RST-1** K-codes followed by one **RST-2** K-code (see [Table 5.11, "Hard Reset Ordered Set"](#)).

Table 5.11 Hard Reset Ordered Set

K-Code Number	K-Code in Code Table
1	RST-1
2	RST-1
3	RST-1
4	RST-2

A device **Shall** perform a *Hard Reset* when it receives **Hard Reset Signaling**. After receiving the **Hard Reset Signaling**, the device **Shall** reset as described in [Section 6.8.3, "Hard Reset"](#). If a **Valid Hard Reset** is not detected (see [Table 5.3, "Validation of Ordered Sets"](#)) then the whole transmission **Shall** be **Discarded**.

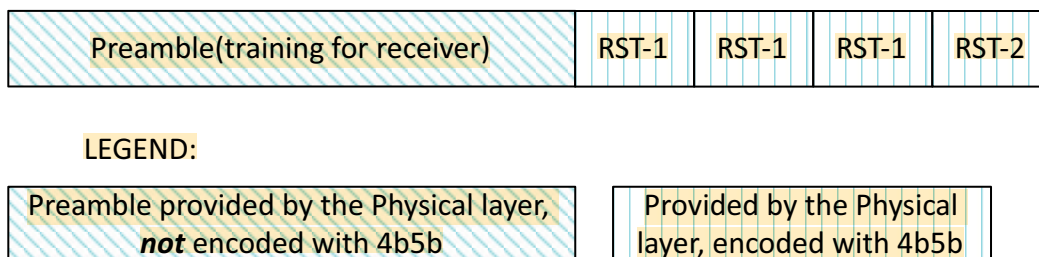
A *Cable Plug* **Shall** perform a *Hard Reset* when it detects **Hard Reset Signaling** being sent between the *Port Partners*. After receiving the **Hard Reset Signaling**, the device **Shall** reset as described in [Section 6.8.3, "Hard Reset"](#).

The procedure for sending **Hard Reset Signaling** **Shall** be as follows:

- If the *PHY Layer* is currently sending a *Message*, the *Message* **Shall** be interrupted by sending an **EOP K-code** and the rest of the *Message* **Discarded**.
- If CC is not *Idle*, wait for it to become *Idle* (see [Section 5.8.6.1, "Definition of Idle"](#)).
- Wait **tInterFrameGap**.
- If CC is still *Idle* send the *Preamble* followed by the 4 K-codes for **Hard Reset Signaling**.
- Disable the channel (i.e., stop sending and receiving), reset the *PHY Layer* and inform the *Protocol Layer* that the *PHY Layer* has been reset.
- Re-enable the channel when requested by the *Protocol Layer*.

[Figure 5.5, "Line format of Hard Reset"](#) shows the line format of **Hard Reset Signaling** which is a *Preamble* followed by the **Hard Reset** Ordered Set.

Figure 5.5 Line format of Hard Reset



5.6.5 Cable Reset

Cable Reset Signaling is an ordered set of bytes sent with the purpose to be recognized by the *PHY* Layer. The **Cable Reset** Signaling ordered set is defined as the following sequence of *K*-codes: *RST-1*, *Sync-1*, *RST-1*, *Sync-3* (see [Table 5.12, "Cable Reset Ordered Set"](#)).

Table 5.12 Cable Reset Ordered Set

K-Code Number	K-Code in Code Table
1	<i>RST-1</i>
2	<i>Sync-1</i>
3	<i>RST-1</i>
4	<i>Sync-3</i>

Cable Reset Signaling **Shall** only be sent by the *DFP*. The **Cable Reset** Ordered Set is used to reset the *Cable Plugs* without the need to *Hard Reset* the *Port Partners*. The state of the *Cable Plug* after the **Cable Reset** Signaling **Shall** be equivalent to power cycling the *Cable Plug*.

[Figure 5.6, "Line format of Cable Reset"](#) shows the line format of **Cable Reset** Signaling which is a *Preamble* followed by the **Cable Reset** Ordered Set.

Figure 5.6 Line format of Cable Reset



LEGEND:



5.7 Collision Avoidance

The *PHY Layer* **Shall** monitor the channel for data transmission and only initiate transmissions when *CC* is *Idle*. If the bus *Idle* condition is present, it **Shall** be considered safe to start a transmission provided the conditions detailed in [Section 5.8.5.4, "Inter-Frame Gap"](#) are met. The bus *Idle* condition **Shall** be checked immediately prior to transmission. If transmission cannot be initiated, then the *Packet* **Shall** be **Discarded**. If the *Packet* is **Discarded** because *CC* is not *Idle*, the *PHY Layer* **Shall** signal to the *Protocol Layer* that it has **Discarded** the *Message* as soon as *CC* becomes *Idle*. See [Section 5.8.6.1, "Definition of Idle"](#) for the definition of *Idle CC*.

In addition, during an *Explicit Contract*, the *PHY Layer* **Shall** control the R_p resistor value to avoid collisions between *Source* and *Sink* transmissions. The *Source* **Shall** set an R_p value corresponding to a current of 3A (**SinkTxOK**) to indicate to the *Sink* that it **May** initiate an AMS. The *Source* **Shall** set an R_p value corresponding to a current of 1.5A (**SinkTxNG**) this **Shall** indicate to the *Sink* that it **Shall Not** initiate an AMS and **Shall** only respond to *Messages* as part of an AMS. See [\[USB Type-C 2.4\] \(USB Type-C\)](#) for details of the corresponding R_p values. During the *Implicit Contract* that precedes an *Explicit Contract* (including *Power Role Swap* and *Fast Role Swap*) the R_p resistor value is used to specify *USB Type-C* current and is not used for *Collision Avoidance*.

[Table 5.13, " \$R_p\$ values used for Collision Avoidance"](#) details the R_p values that **Shall** be used by the *Source* to control *Sink* initiation of an AMS.

Table 5.13 R_p values used for Collision Avoidance

Source R_p	Parameter	Description	Sink Operation	Source Operation
1.5A@5V	SinkTxNG	<i>Sink</i> Transmit "No Go,"	The <i>Sink</i> Shall Not initiate an AMS once $t_{SinkDelay}$ has elapsed after SinkTxNG is asserted.	<i>Source</i> can initiate an AMS t_{SinkTx} after setting R_p to this value.
3A@5V	SinkTxOK	<i>Sink</i> Transmit "Ok"	<i>Sink</i> can initiate an AMS.	<i>Source</i> cannot initiate an AMS while it has this value set.

See also [Section 6.6.16, "Collision Avoidance Timers"](#) and [Section 6.10, "Collision Avoidance"](#).

5.8 Bi-phase Mark Coding (BMC) Signaling Scheme

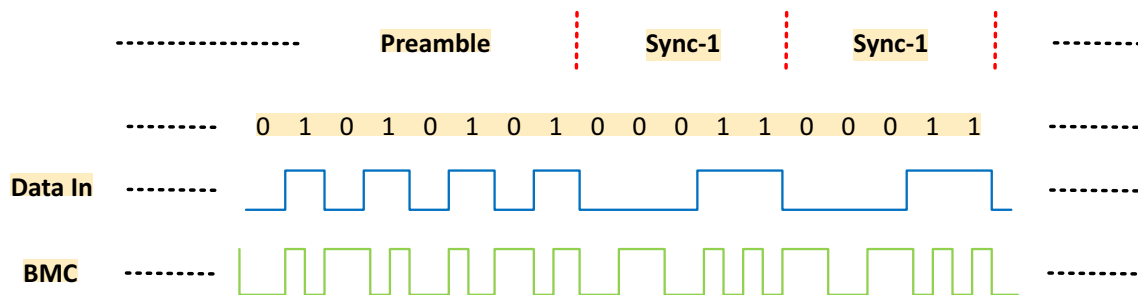
Bi-phase Mark Coding (BMC) is the *PHY Layer Signaling Scheme* for carrying USB Power Delivery Messages. This encoding assumes a dedicated DC connection, **over** the CC wire, which is used for sending PD Messages.

Bi-phase Mark Coding is a version of Manchester coding (see [IEC 60958-1]). In *BMC*, there is a transition at the start of every bit time (*UI*) and there is a second transition in the middle of the *UI* when a 1 is transmitted. *BMC* is effectively DC balanced, (each 1 is DC balanced and two successive **zeros** are DC balanced, regardless of the number of intervening 1's). It has bounded disparity (limited to 1 bit over an arbitrary *Packet*, so a very low DC level).

Figure 5.7, "BMC Example" illustrates *Bi-phase Mark Coding*. This example shows the transition from a *Preamble* to the *Sync-1* K-codes of the *SOP* Ordered Set at the start of a *Message*.

Note: Other K-codes can occur after the *Preamble* for Signaling such as *Hard Reset* and *Cable Reset*.

Figure 5.7 BMC Example



5.8.1 Encoding and signaling

BMC uses DC coupled baseband *Signaling* on CC. *Figure 5.8, "BMC Transmitter Block Diagram"* shows a block diagram for a Transmitter and *Figure 5.9, "BMC Receiver Block Diagram"* shows a block diagram for the corresponding Receiver.

Figure 5.8 BMC Transmitter Block Diagram

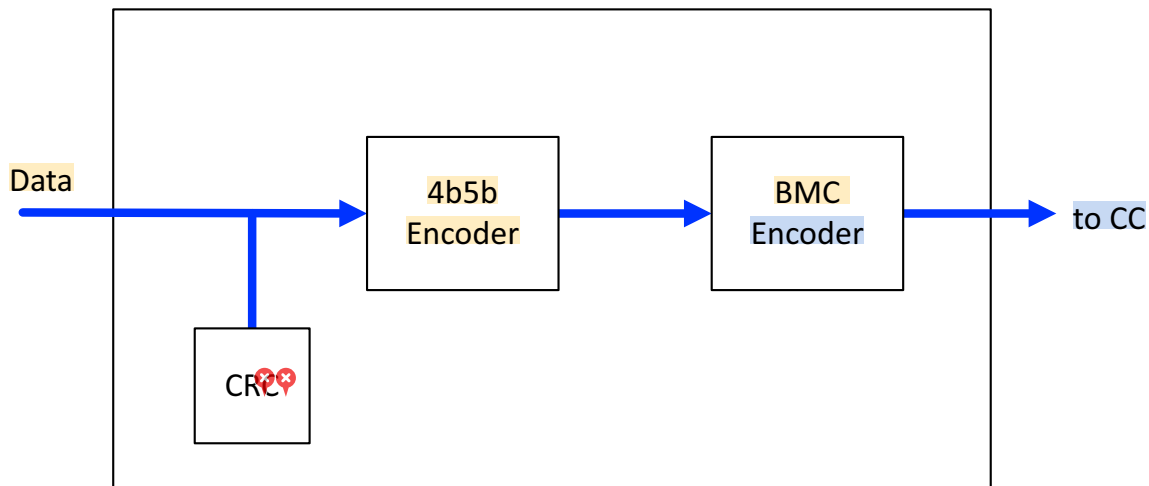
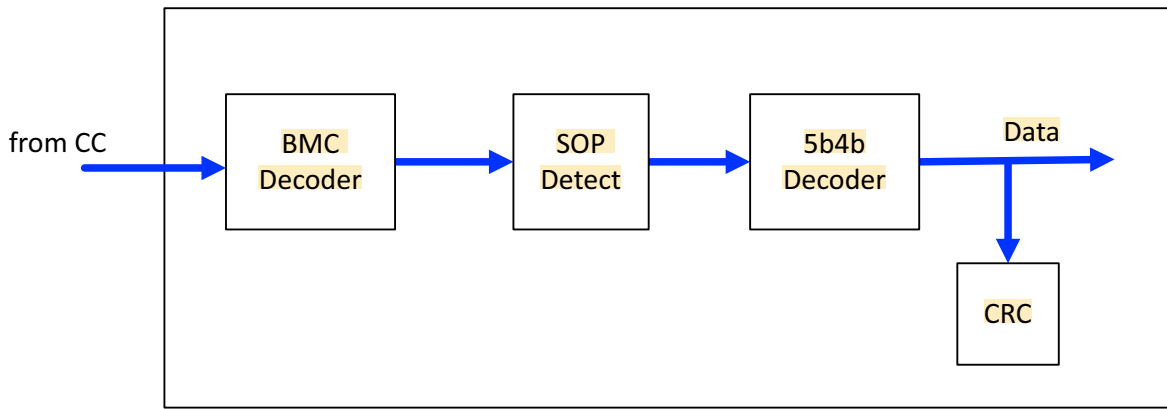


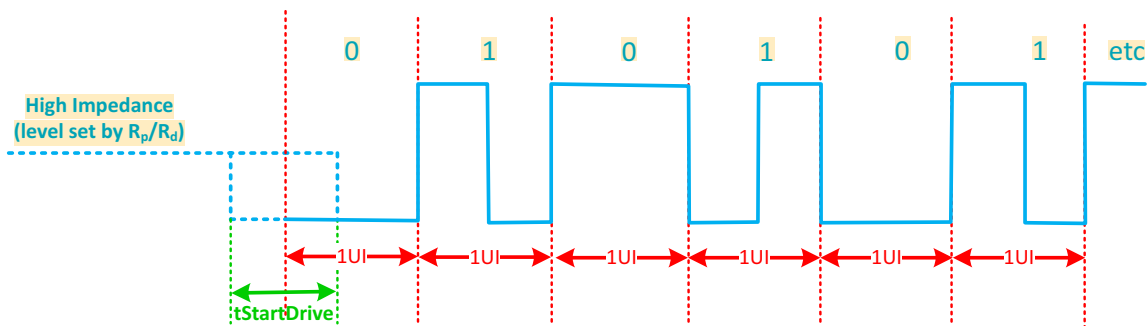
Figure 5.9 BMC Receiver Block Diagram



The USB PD baseband signal **shall** be driven on the CC wire with a tristate driver that **shall** cause a **vSwing** swing on CC. The tristate driver is slew rate limited (see min rise/fall time in [Section 5.8.5, "BMC Transmitter Specifications"](#)) to limit coupling to D+/D- and to other signal lines in the *USB Type-C* fully featured cables (see [\[USB Type-C 2.4\]](#)). This slew rate limiting can be performed either with driver design or an RC filter on the driver output.

When sending the *Preamble*, the transmitter **shall** start by transmitting a low level. The receiver **shall** tolerate the loss of the first edge. The transmitter **may** vary the start of the *Preamble* by **tStartDrive** min (see [Figure 5.10, "BMC Encoded Start of Preamble"](#)).

Figure 5.10 BMC Encoded Start of Preamble



The transmitter **shall** terminate the final bit of the *Frame* by an edge (the "trailing edge") to help ensure that the receiver clocks the final bit. If the trailing edge results in the transmitter driving CC low (i.e., the final half-UI of the *Frame* is high, see [Figure 5.11, "Transmitting or Receiving BMC Encoded Frame Terminated by Zero with High-to-Low Last Transition"](#) and [Figure 5.12, "Transmitting or Receiving BMC Encoded Frame Terminated by One with High-to-Low Last Transition"](#)), then the transmitter:

- **shall** continue to drive CC low for **tHoldLowBMC**.
- **should** release CC to high impedance as soon as possible after min **tHoldLowBMC** and **shall** release CC by max **tEndDriveBMC**.

[Figure 5.11, "Transmitting or Receiving BMC Encoded Frame Terminated by Zero with High-to-Low Last Transition"](#) illustrates the end of a BMC encoded *Frame* with an encoded zero for which the final bit of the *Frame* is terminated by a high to low transition. [Figure 5.12, "Transmitting or Receiving BMC Encoded Frame Terminated by One with High-to-Low Last Transition"](#) illustrates the end of a BMC Encoded *Frame* with an encoded one for which the final bit of the *Frame* is terminated by a high to low transition. Both figures also illustrate the **tInterFrameGap** timing requirement before the start of the next *Frame* when the *Port* has either been transmitting or receiving the previous *Frame* (see [Section 5.8.5.4, "Inter-Frame Gap"](#)).

Figure 5.11 Transmitting or Receiving BMC Encoded Frame Terminated by Zero with High-to-Low Last Transition

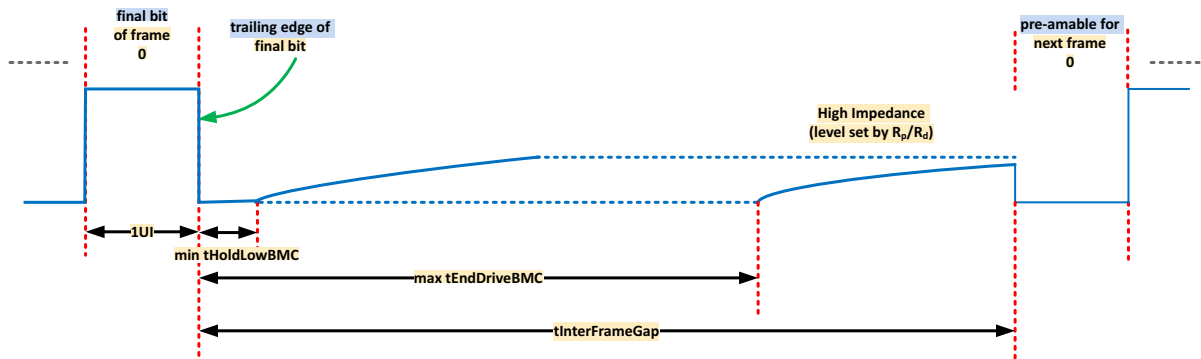
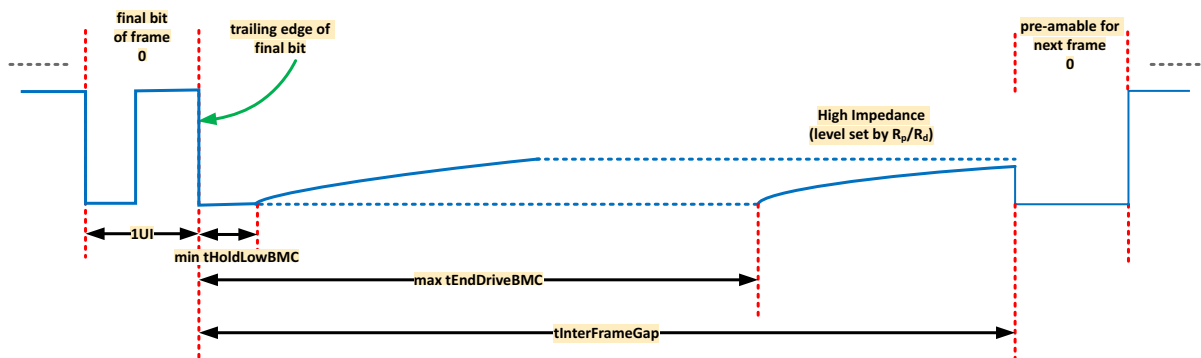


Figure 5.12 Transmitting or Receiving BMC Encoded Frame Terminated by One with High-to-Low Last Transition



If the trailing edge results in the transmitter driving CC high (i.e., the final half-UI of the Frame is low, see [Figure 5.13, "Transmitting or Receiving BMC Encoded Frame Terminated by Zero with Low to High Last Transition"](#) and [Figure 5.14, "Transmitting or Receiving BMC Encoded Frame Terminated by One with Low to High Last Transition"](#)), then the transmitter:

- **Shall** continue to drive CC high for 1 UI.
 - Then **Shall** drive CC low for **tHoldLowBMC**.
- **Should** release CC to high impedance as soon as possible after min **tHoldLowBMC** and **Shall** release CC by max **tEndDriveBMC**.

[Figure 5.13, "Transmitting or Receiving BMC Encoded Frame Terminated by Zero with Low to High Last Transition"](#) illustrates the ending of a BMC encoded Frame that ends with an encoded zero for which the final bit of the Frame is terminated by a low to high transition. [Figure 5.14, "Transmitting or Receiving BMC Encoded Frame Terminated by One with Low to High Last Transition"](#) illustrates the ending of a BMC encoded Frame that ends with an encoded one for which the final bit of the Frame is terminated by a low to high transition. Both figures also illustrate the **tInterFrameGap** timing requirement before the start of the next Frame when the Port has either been transmitting or receiving the previous Frame (see [Section 5.8.5.4, "Inter-Frame Gap"](#)).

Figure 5.13 Transmitting or Receiving BMC Encoded Frame Terminated by Zero with Low to High Last Transition

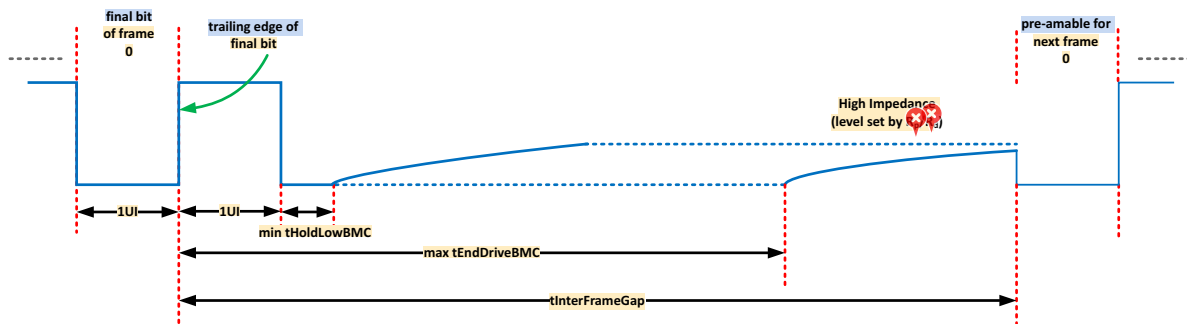
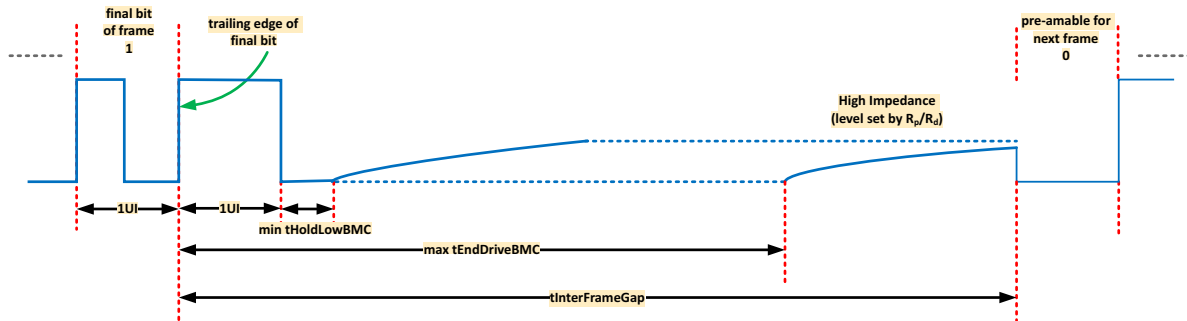


Figure 5.14 Transmitting or Receiving BMC Encoded Frame Terminated by One with Low to High Last Transition



Note: There is no requirement to maintain a timing phase relationship between back-to-back Packets.

5.8.2 Transmit and Receive Masks

5.8.2.1 Transmit Masks

The transmitted signal **Shall Not** violate the masks defined in [Figure 5.15, "BMC Tx 'ONE' Mask"](#), [Figure 5.16, "BMC Tx 'ZERO' Mask"](#), [Table 5.14, "BMC Tx Mask Definition, X Values"](#) and [Table 5.15, "BMC Tx Mask Definition, Y Values"](#) at the output of a load equivalent to the cable model and receiver load model described in [Section 5.8.3, "Transmitter Load Model"](#). The masks apply to the full range of R_p/R_d values as defined in [\[USB Type-C 2.4\]](#).

Note: The measurement of the transmitter does not need to accommodate a change in signal offset due to the ground offset when current is flowing in the cable.

The transmitted signal **Shall** have a rise time no faster than t_{Rise} . The transmitted signal **Shall** have a fall time no faster than t_{Fall} . The maximum limits on the rise and fall times are enforced by the Tx inner masks.

Figure 5.15 BMC Tx 'ONE' Mask

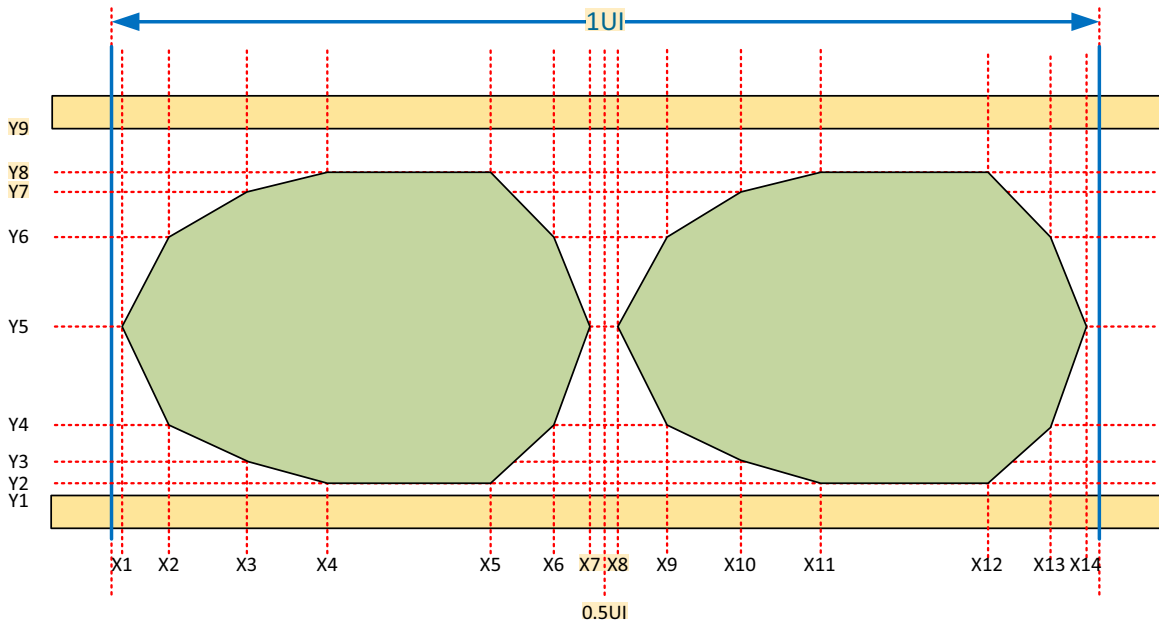


Figure 5.16 BMC Tx 'ZERO' Mask

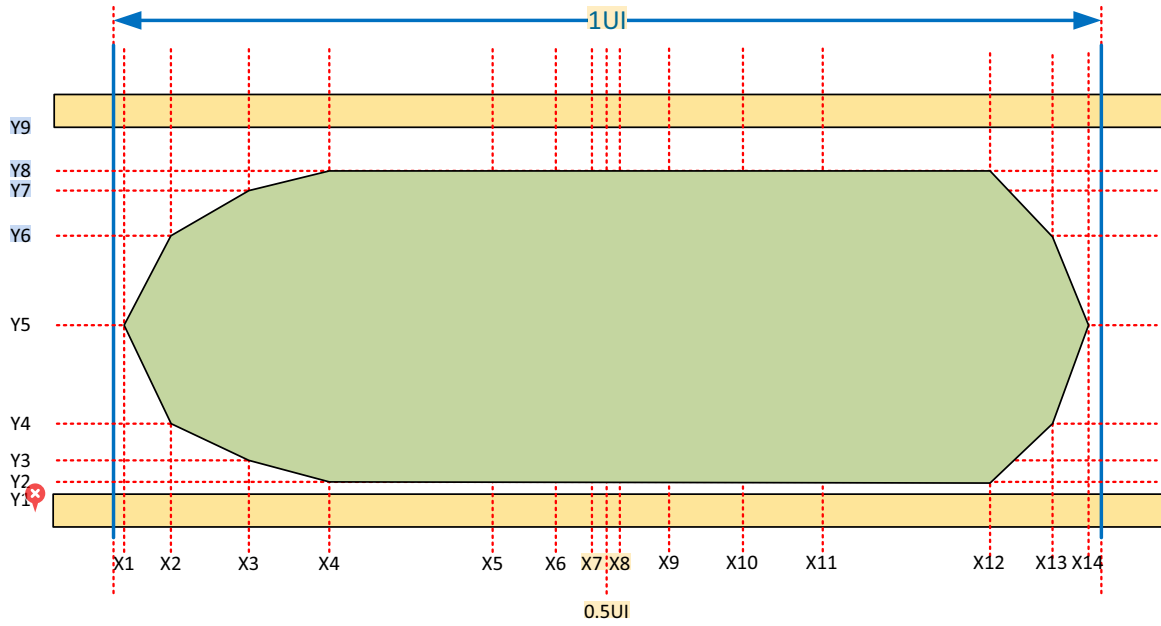


Table 5.14 BMC Tx Mask Definition, X Values

Name	Description	Value	Units
X1Tx	Left Edge of Mask	0.015	UI
X2Tx	see figure	0.07	UI
X3Tx	see figure	0.15	UI
X4Tx	see figure	0.25	UI
X5Tx	see figure	0.35	UI
X6Tx	see figure	0.43	UI
X7Tx	see figure	0.485	UI
X8Tx	see figure	0.515	UI
X9Tx	see figure	0.57	UI
X10Tx	see figure	0.65	UI
X11Tx	see figure	0.75	UI
X12Tx	see figure	0.85	UI
X13Tx	see figure	0.93	UI
X14Tx	Right Edge of Mask	0.985	UI

Table 5.15 BMC Tx Mask Definition, Y Values

Name	Description	Value	Units
Y1Tx	Lower bound of Outer mask	-0.075	V
Y2Tx	Lower bound of inner mask	0.075	V _×
Y3Tx	see figure	0.15	V
Y4Tx	see figure	0.325	V
Y5Tx	Inner mask vertical midpoint	0.5625	V _×
Y6Tx	see figure	0.8	V
Y7Tx	see figure	0.975	V
Y8Tx	see figure	1.04	V
Y9Tx	Upper Bound of Outer mask	1.2	V _×

5.8.2.2 Receive Masks

A *Source* using the *BMC Signaling Scheme* **Shall** be capable of receiving a signal that complies with the mask when sourcing power as defined in [Figure 5.17, "BMC Rx 'ONE' Mask when Sourcing Power"](#), [Figure 5.18, "BMC Rx 'ZERO' Mask when Sourcing Power"](#) and [Table 5.16, "BMC Rx Mask Definition"](#). The *Source* Rx mask is bounded by sweeping a Tx mask compliant signal, with added **vNoiseActive** between power neutral and *Source* offsets.

A *Consumer* using the *BMC Signaling Scheme* **Shall** be capable of receiving a signal that complies with the mask when sinking power as defined in [Figure 5.21, "BMC Rx 'ONE' Mask when Sinking Power"](#), [Figure 5.22, "BMC Rx 'ZERO' Mask when Sinking Power"](#) and [Table 5.16, "BMC Rx Mask Definition"](#). The *Consumer* Rx mask is bounded by sweeping a Tx mask compliant signal, with added **vNoiseActive** between power neutral and *Consumer* offsets.

Every product using the *BMC Signaling Scheme* **Shall** be capable of receiving a signal that complies with the mask when power neutral as defined in [Figure 5.19, "BMC Rx 'ONE' Mask when Power neutral"](#), [Figure 5.20, "BMC Rx 'ZERO' Mask when Power neutral"](#) and [Table 5.16, "BMC Rx Mask Definition"](#).

Dual-Role Power Devices **Shall** meet the receiver requirements for a *Source* when providing power during any transmission using the *BMC Signaling Scheme* or a *Sink* when consuming power during any transmission using the *BMC Signaling Scheme*.

Cable Plugs **Shall** meet the receiver requirements for both a *Source* and a *Sink* during any transmission using the *BMC Signaling Scheme*.

The parameters used in the masks are specified to be appropriate to either edge triggered or oversampling receiver implementations.

The masks are defined for 'ONE' and 'ZERO' separately as *BMC* enforces a transition at the midpoint of the unit interval while a 'ONE' is transmitted.

The Rx masks are defined to bound the Rx noise after the Rx bandwidth limiting filter with the time constant **tRxFilter** has been applied.

The boundaries of Rx outer mask, **Y1Rx** and **Y5Rx**, are specified according to **vSwing** max and accommodate half of **vNoiseActive** from cable noise coupling and the signal offset **vIRDropGNDC** due to the ground offset when current is flowing in the cable.

The vertical dimension of the Rx inner mask, **Y4Rx** - **Y2Rx**, for power neutral is derived by reducing the vertical dimension of the Tx inner mask, **Y7Tx** - **Y3Tx**, at time location **X3Tx** by **vNoiseActive** to account for cable noise coupling. The received signal is composed of a waveform compliant to the Tx mask plus **vNoiseActive**.

The vertical dimension of the Rx inner mask for sourcing power is derived by reducing the vertical dimension of the Tx inner mask by **vNoiseActive** and **vIRDropGNDC** to account for both cable noise coupling and signal DC offset.

The received signal is composed of a waveform compliant to the Tx mask plus the maximum value of $v_{NoiseActive}$ plus $v_{IRDropGNDC}$ where the $v_{IRDropGNDC}$ value transitions between the minimum and the maximum values as allowed in this spec.

The vertical dimension of the Rx inner mask for sinking power is derived by reducing the vertical dimension of the Tx inner mask by $v_{NoiseActive}$ max and $v_{IRDropGNDC}$ max for account for both cable noise coupling and signal DC offset. The received signal is composed of a waveform compliant to the Tx mask plus the maximum value of $v_{NoiseActive}$ plus $v_{IRDropGNDC}$ where the $v_{IRDropGNDC}$ value transitions between the minimum and the maximum values as allowed in this spec.

The center line of the Rx inner mask, Y_{3Rx} , is at half of the nominal v_{Swing} for power neutral, and is shifted up by half of $v_{IRDropGNDC}$ max for sourcing power and is shifted down by half of $v_{IRDropGNDC}$ max for sinking power.

The receiver sensitivity **shall** be set such that the receiver does not treat noise on an undriven signal path as an incoming signal. Signal amplitudes below $v_{NoiseIdle}$ max **shall** be treated as noise when BMC is *Idle*.

Figure 5.17 BMC Rx 'ONE' Mask when Sourcing Power

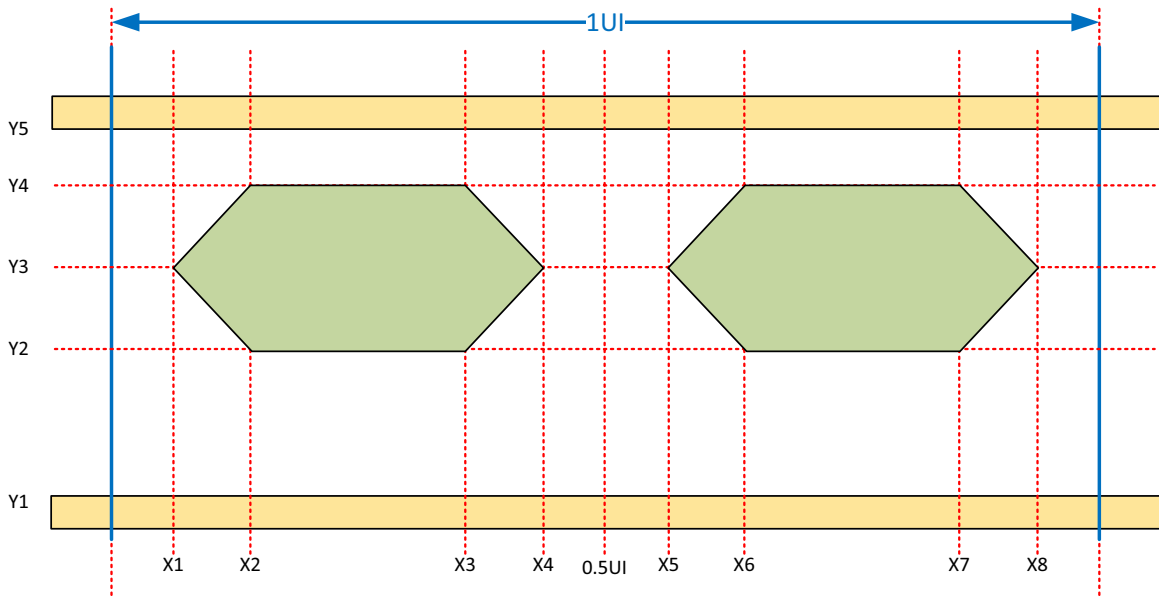


Figure 5.18 BMC Rx 'ZERO' Mask when Sourcing Power

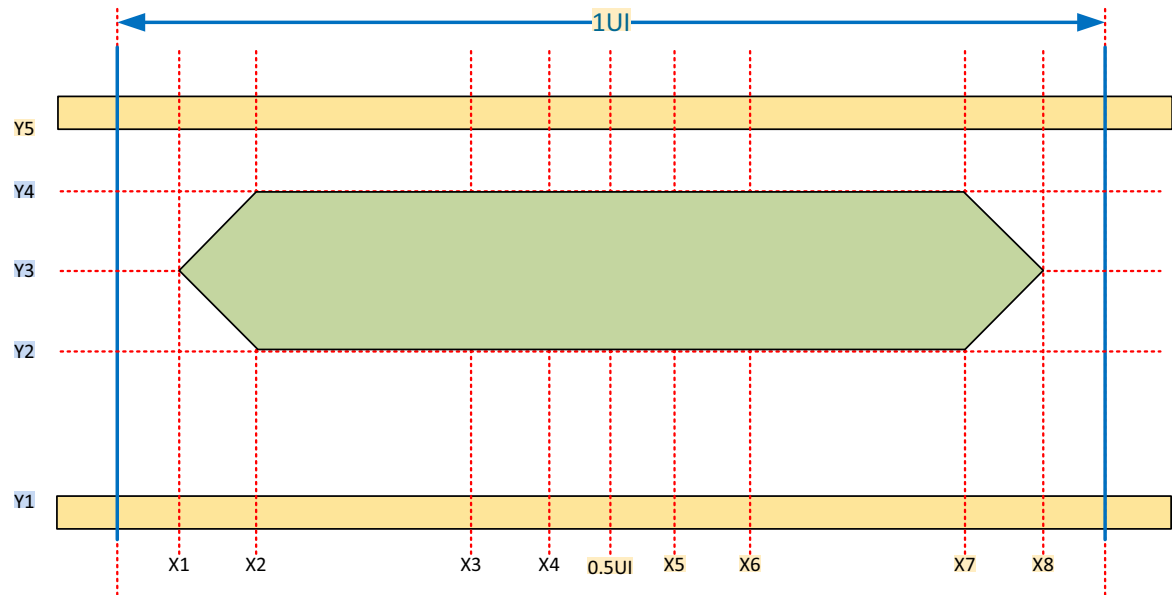


Figure 5.19 BMC Rx 'ONE' Mask when Power neutral

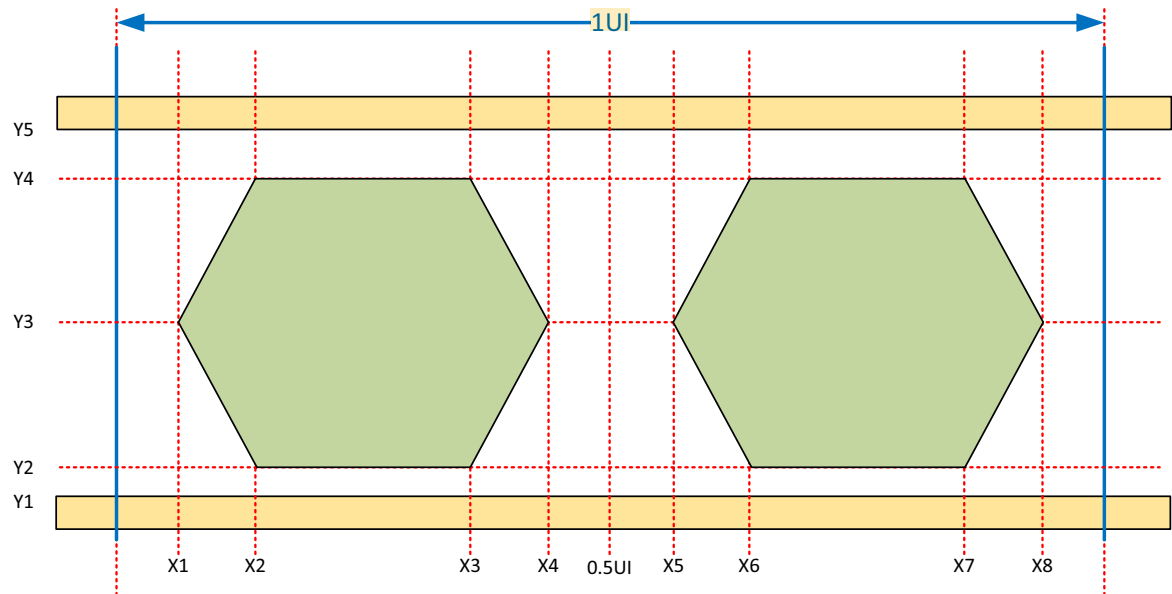


Figure 5.20 BMC Rx 'ZERO' Mask when Power neutral

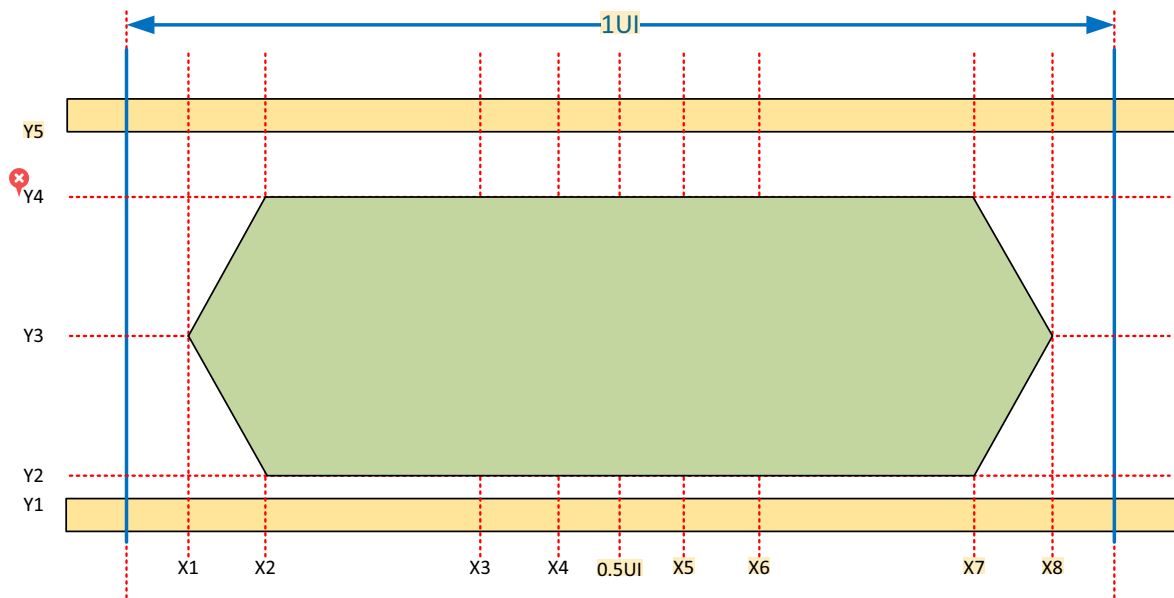


Figure 5.21 BMC Rx 'ONE' Mask when Sinking Power

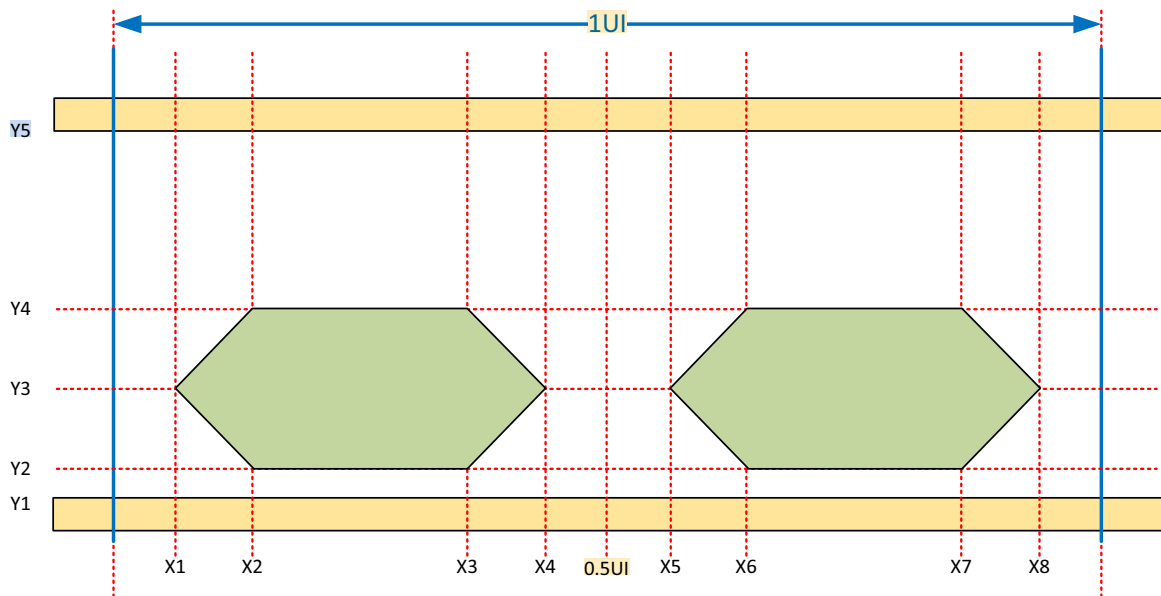


Figure 5.22 BMC Rx 'ZERO' Mask when Sinking Power

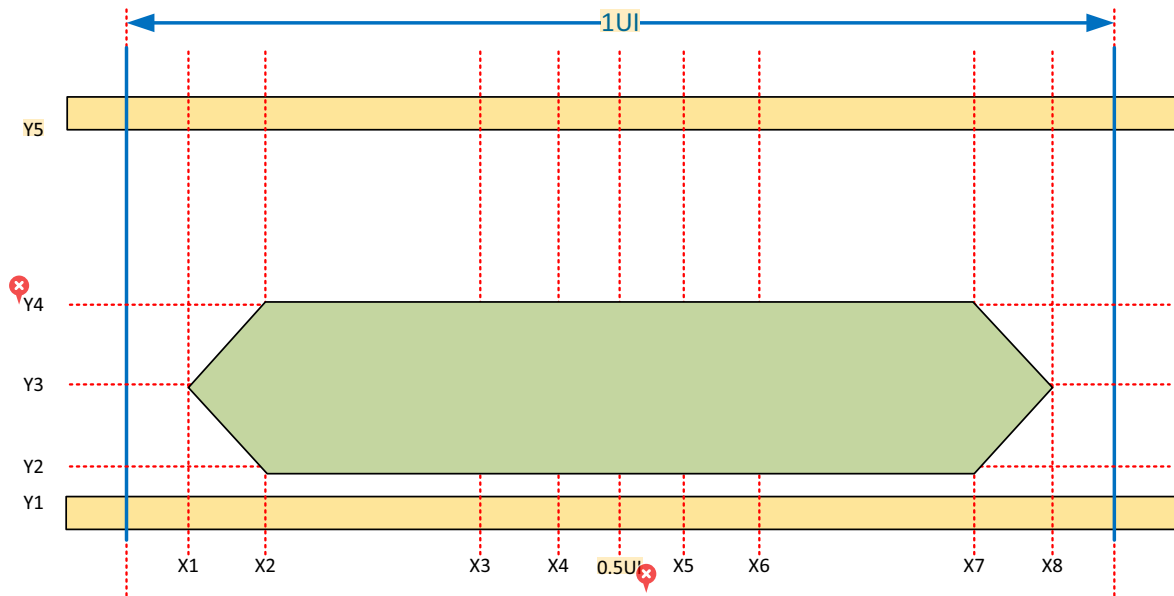


Table 5.16 BMC Rx Mask Definition

Name	Description	Value	Units
X1Rx	Left Edge of Mask	0.07	UI
X2Rx	Top Edge of Mask	0.15	UI
X3Rx	See figure	0.35	UI
X4Rx	See figure	0.43	UI
X5Rx	See figure	0.57	UI
X6Rx	See figure	0.65	UI
X7Rx	See figure	0.85	UI
X8Rx	See figure	0.93	UI
Y1Rx	Lower bound of Outer Mask	-0.3325	V
Y2Rx	Lower Bound of Inner Mask	Y3Rx - 0.205 when sourcing power ¹ or sinking power ¹ . Y3Rx - 0.33 when power neutral ¹ .	V
Y3Rx	Center line of Inner Mask	0.6875 Sourcing Power ¹ . 0.5625 Power Neutral ¹ . 0.4375 Sinking Power ¹ .	V
Y4Rx	Upper bound of Inner mask	Y3Rx + 0.205 when sourcing power ¹ or sinking power ¹ . Y3Rx + 0.33 when power neutral ¹ .	V
Y5Rx	Upper bound of the Outer mask	1.5325	V

1) The position of the center line of the Inner Mask is dependent on whether the receiver is Sourcing or Sinking power or is Power Neutral (see earlier in this section).

5.8.3 Transmitter Load Model

The transmitter load model **shall** be equivalent to the circuit outlined in [Figure 5.23, "Transmitter Load Model for BMC Tx from a Source"](#) for a *Source* and [Figure 5.24, "Transmitter Load Model for BMC Tx from a Sink"](#) for a *Sink*. It is formed by the concatenation of a cable load model and a receiver load model. See [\[USB Type-C 2.4\]](#) for details of the R_p and R_d resistors.

Note: The parameters $zCable_CC$, $tCableDelay_CC$ and $cCablePlug_CC$ are defined in [\[USB Type-C 2.4\]](#).

Figure 5.23 Transmitter Load Model for BMC Tx from a Source

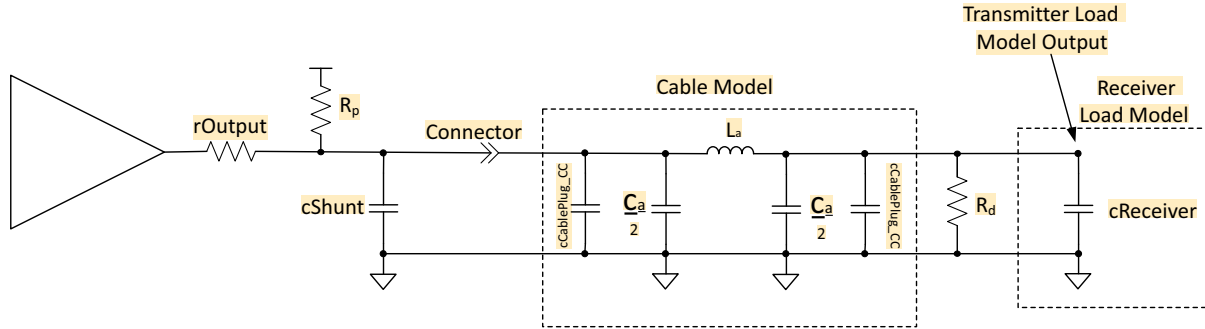
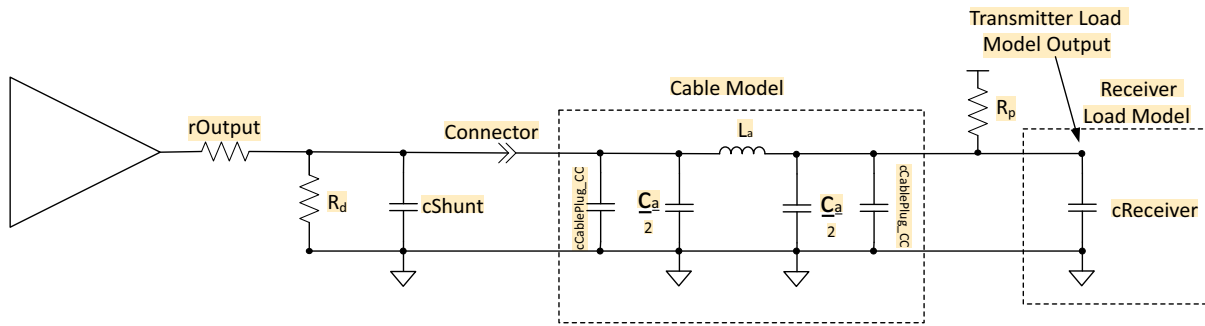


Figure 5.24 Transmitter Load Model for BMC Tx from a Sink



The transmitter system components $rOutput$ and $cShunt$ are illustrated for **Informative** purposes, and do not form part of the transmitter load model. See [Section 5.8.5, "BMC Transmitter Specifications"](#) for a description of the transmitter system design.

The value of the modeled cable inductance, L_a , (in nH) **shall** be calculated from the following formula:

$$L_a = tCableDelay_CC_{max} * zCable_CC_{min}$$

$tCableDelay_CC$ is the modeled signal propagation delay through the cable, and $zCable_CC$ is the modeled cable impedance.

The modeled cable inductance is 640nH for a cable with $zCable_CC_{min} = 32\Omega$ and $tCableDelay_CC_{max} = 20ns$.

The value of the modeled cable capacitance, C_a , (in pF) **shall** be calculated from the following formula:

$$C_a = tCableDelay_CC_{max} / zCable_CC_{min}$$

The modeled cable capacitance is $C_a = 625pF$ for a cable with $zCable_CC_{min} = 32\Omega$ and $tCableDelay_CC_{max} = 20ns$. Therefore, $C_a/2 = 312.5pF$.

$cCablePlug_CC$ models the capacitance of the plug at each end of the cable. $cReceiver$ models the capacitance of the receiver. The maximum values **shall** be used in each case.

Note: The transmitter load model assumes that there are no other return currents on the ground path.

5.8.4 BMC Common specifications

This section defines the common receiver and transmitter requirements.

5.8.4.1 BMC Common Parameters

The electrical requirements specified in [Table 5.17, "BMC Common Normative Requirements"](#) **shall** apply to both the transmitter and receiver.

Table 5.17 BMC Common Normative Requirements

Name	Description	Min	Nom	Max	Units	Comment
<i>fBitRate</i>	Bit rate	270	300	330	Kbps	
<i>tUnitInterval</i>	Unit Interval 	3.03		3.70	μs	$1/f\textit{BitRate}$
1) Denotes the time to transmit an unencoded data bit, not the shortest high or low times on the wire after encoding with BMC. A single data bit cell has duration of 1UI, but a data bit cell with value 1 will contain a centrally placed 01 or 10 transition in addition to the transition at the start of the cell.						

5.8.5

BMC Transmitter Specifications

The transmitter **shall** meet the specifications defined in [Table 5.18, "BMC Transmitter Normative Requirements"](#).

Table 5.18 BMC Transmitter Normative Requirements

Name	Description	Min	Nom	Max	Units	Comment
pBitRate	Maximum difference between the bit-rate during the part of the <i>Packet</i> following the <i>Preamble</i> and the reference bit-rate.			0.25	%	The reference bit rate is the average bit rate of the last 32 bits of the <i>Preamble</i> .
rFRSwapTx	<i>Fast Role Swap Request</i> transmit driver resistance (excluding cable resistance)			5	Ω	Maximum driver resistance of a <i>Fast Role Swap Request</i> transmitter. Assumes a worst case cable resistance of 15Ω as defined in [USB Type-C 2.4] . Note: Based on this value the maximum combined driver and cable resistance of a <i>Fast Role Swap Request</i> transmitter is 20Ω .
tEndDriveBMC	Time to cease driving the line after the end of the last bit of the <i>Frame</i> .			23	μs	Min value is limited by tHoldLowBMC .
tFall	Fall Time	300			ns	10% and 90% amplitude points, minimum is under an unloaded condition.
tHoldLowBMC	Time to cease driving the line after the final high-to-low transition.	1			μs	Max value is limited by tEndDriveBMC .
tInterFrameGap	Time from the end of last bit of a <i>Frame</i> until the start of the first bit of the next <i>Preamble</i> .	25			μs	
tFRSwapTx	<i>Fast Role Swap Request</i> transmit duration	60		120	μs	<i>Fast Role Swap Request</i> is indicated from the <i>Initial Source</i> to the <i>Initial Sink</i> by driving <i>CC</i> low for this time.
tRise	Rise time	300			ns	10% and 90% amplitude points, minimum is under an unloaded condition.
tStartDrive	Time before the start of the first bit of the <i>Preamble</i> when the transmitter shall start driving the line.	-1		1	μs	
vSwing	Voltage Swing	1.05	1.125	1.2	V	Applies to both no load condition and under the load condition specified in Section 5.8.3, "Transmitter Load Model" .
zDriver	Transmitter output impedance	33		75	Ω	<i>Source</i> output impedance at the Nyquist frequency of [USB 2.0] low speed (750 kHz) while the <i>Source</i> is driving the <i>CC</i> line.

5.8.5.1 Capacitance when not transmitting

cReceiver is the capacitance that a *DFP* or *UFP* **Shall** present on the *CC* line when the *DFP* or *UFP's* receiver is not transmitting on the line. The transmitter **May** have more capacitance than **cReceiver** while driving the *CC* line, but **Shall** meet the waveform mask requirements. Once transmission is complete, the transmitter **Shall** disengage capacitance in excess of **cReceiver** from the *CC* wire within **tInterFrameGap**.

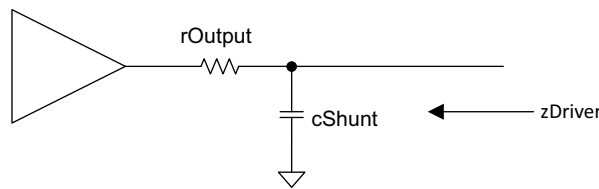
5.8.5.2 Source Output Impedance

Source output impedance **zDriver** is determined by the driver resistance and the shunt capacitance of the *Source* and is hence a frequency dependent term. **zDriver** impacts the noise ingress in the cable. It is specified such that the noise at the Receiver is bounded.

zDriver is defined by the following equation:

$$zDriver = rOutput / (1 + s * rOutput * cShunt)$$

Figure 5.25 Transmitter diagram illustrating zDriver



cShunt **Shall** Not cause a violation of **cReceiver** when not transmitting.

5.8.5.3 Bit Rate Drift

Limits on the drift in **fBitRate** are set to help low-complexity receiver implementations.

fBitRate is the reciprocal of the average bit duration from the previous 32 bits at a given portion of the *Packet*. The change in **fBitRate** during a *Packet* **Shall** be less than **pBitRate**. The reference bit rate (refBitRate) is the average **fBitRate** over the last 32 bits of the *Preamble*. **fBitRate** throughout the *Packet*, including the *EOP*, **Shall** be within **pBitRate** of refBitRate. **pBitRate** is expressed as a percentage:

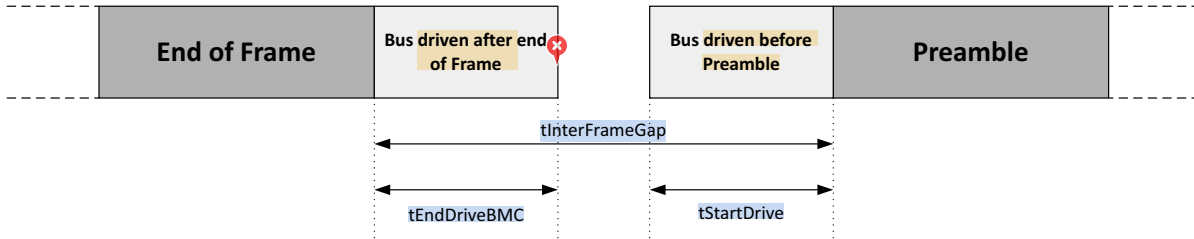
$$pBitRate = | fBitRate - refBitRate | / refBitRate \times 100\%$$

The transmitter **Shall** have the same **pBitRate** for all *Packet* types. The *BIST Carrier Mode* and *Bit Stream* signals are continuous signals without a *Payload*. When checking **pBitRate** any set of 1044 bits (20 bit **SOP** followed by 1024 PRBS bits) within a continuous signal **May** be considered as the part of the *Packet* following the *Preamble* and the 32 preceding bits considered to be the last 32 bits of the *Preamble* used to compute refBitRate.

5.8.5.4 Inter-Frame Gap

Figure 5.26, "Inter-Frame Gap Timings" illustrates the inter-Frame gap timings.

Figure 5.26 Inter-Frame Gap Timings



The transmitter **shall** drive the bus for no longer than $t_{\text{EndDriveBMC}}$ after transmitting the final bit of the *Frame*.

Before starting to transmit the next *Frame's Preamble* the transmitter of the next *Frame* **shall** ensure that it waits for $t_{\text{InterFrameGap}}$ after either:

- Transmitting the previous *Frame*, for example sending the next *Message* in an *AMS* immediately after having sent a *GoodCRC Message*, or
- Receiving the previous *Frame*, for example when responding to a received *Message* with a *GoodCRC Message*, or
- Observing an *Idle* condition on *CC* (see [Section 5.7, "Collision Avoidance"](#)). In this case the *Port* is waiting to initiate an *AMS* observes *Idle* (see [Section 5.8.6.1, "Definition of Idle"](#)) and then waits $t_{\text{InterFrameGap}}$ before transmitting the *Frame*. See also [Section 5.7, "Collision Avoidance"](#) for details on when an *AMS* can be initiated.

Note: The transmitter is also required to verify a bus *Idle* condition immediately prior to starting transmission of the next *Frame* (see [Section 5.8.6.1, "Definition of Idle"](#)).

The transmitter of the next *Frame* **may** vary the start of the *Preamble* by $t_{\text{StartDrive}}$ (see [Section 5.8.1, "Encoding and signaling"](#)).

See also [Section 5.8.1, "Encoding and signaling"](#) for figures detailing the timings relating to transmitting, receiving, and observing *Idle* in relating to *Frames*.

5.8.5.5 Shorting of Transmitter Output

A Transmitter in a *Port* or *Cable Plug* **shall** tolerate having its output be shorted to ground for t_{FRSwapTx} max. This is due to the potential for *Fast Role Swap* to be signaled while the Transmitter is in the process of transmitting (see [Section 5.8.5.6, "Fast Role Swap Transmission"](#)).

5.8.5.6 Fast Role Swap Transmission

The *Fast Role Swap* process is intended for use by a *PDUSB Hub* that presently has an external supply and is providing power both through its downstream *Ports* to *USB Devices* and upstream to a *USB Host* such as a laptop. On removal of the external wall supply *Fast Role Swap* enables a *VBUS* supply to be maintained by allowing the *USB Host* to apply *vSafe5V* when it sees *VBUS* droop below *vSafe5V* after having detected *Fast Role Swap Signaling*. The *Fast Role Swap AMS* is then used to correctly assign *Source/Sink Power Roles* and configure the R_p/R_d resistors (see [Section 8.3.2.8, "Fast Role Swap"](#)).

The *Initial Source* **Shall** signal a *Fast Role Swap Request* by driving *CC* to ground with a resistance of less than *rFRSwapTx* for *tFRSwapTx*. The *Initial Source* **Shall** only send a *Fast Role Swap Request* when it has an *Explicit Contract*. The *Initial Source* **May** send a *Fast Role Swap Request* even if it has not yet had its *Sink Capabilities* queried by the *Initial Sink*. On transmission of the *Fast Role Swap Request* any pending *Messages* **Shall** be **Discarded** (see [Section 6.12.2.2.1, "Common Protocol Layer Message Transmission State Diagram"](#)).

The *Fast Role Swap Signaling* **May** override any active transmissions.

Since the *Initial Sink's* response to the *Fast Role Swap* signal is to send an *FR_Swap* Message, the *Initial Source* **Shall** ensure R_p is set to *SinkTxOK* once the *Fast Role Swap Signaling* is complete.

5.8.6 BMC Receiver Specifications

The receiver **shall** meet the specifications defined in [Table 5.19, "BMC Receiver Normative Requirements"](#).

Table 5.19 BMC Receiver Normative Requirements

Name	Description	Min	Nom	Max	Units	Comment
cReceiver	CC receiver capacitance	200		600	pF	The DFP or UFP system shall have capacitance within this range when not transmitting on the line.
nBER	Bit error rate, S/N = 25 dB			10 ⁻⁶		
nTransitionCount	Transitions for signal detect	3				Number of transitions to be detected to declare bus non-Idle.
tFRSwapRx	Fast Role Swap Request detection time	30		50	μs	A Fast Role Swap Request results in the receiver detecting a signal low for at least this amount of time.
tRxFilter	Rx bandwidth limiting filter (digital or analog)	100			ns	Time constant of a single pole filter to limit broad-band noise ingress ¹ .
tTransitionWindow	Time window for detecting non-Idle	12		20	μs	
vFRSwapCableTx	Fast Role Swap Request voltage detection threshold	490	520	550	mV	The Fast Role Swap Request must be below this voltage threshold to be detected.
vIRDropGNDC	Cable Ground IR Drop			250	mV	As specified in [USB Type-C 2.4] .
vNoiseActive	Noise amplitude when BMC is active.			165	mV	Peak-to-peak noise from VBUS, [USB 2.0] and SBU lines after the Rx bandwidth limiting filter with the time constant tRxFilter has been applied.
vNoiseIdle	Noise amplitude when BMC is Idle.			300	mV	Peak-to-peak noise from VBUS, [USB 2.0] and SBU lines after the Rx bandwidth limiting filter with the time constant tRxFilter has been applied.
zBmcRx	Receiver Input Impedance	1			MΩ	

1) Broad-band noise ingress is due to coupling in the cable interconnect.

5.8.6.1 Definition of Idle

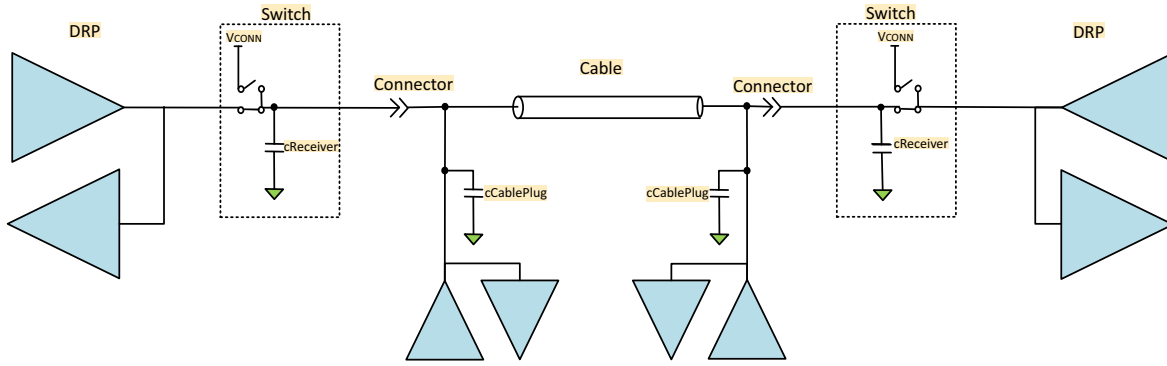
BMC Collision Avoidance is performed by the detection of signal transitions at the receiver. Detection is active when **nTransitionCount** transitions occur at the receiver within a time window of **tTransitionWindow**. After waiting **tTransitionWindow** without detecting **nTransitionCount** transitions the bus **shall** be declared **Idle**.

Refer to [Section 5.8.5.4, "Inter-Frame Gap"](#) for details of when transmissions **may** start.

* The *BMC Signaling Scheme* is suitable for use in *Multi-Drop* configurations containing one or two *BMC Multi-Drop* transceivers connected to the *CC* wire, where one or both ends of a cable contains a *Multi-Drop* transceiver. In this specification the location of the *Multi-Drop* transceiver is referred to as the *Cable Plug*.

Figure 5.27, "Example Multi-Drop Configuration showing two DRPs" below illustrates a typical *Multi-Drop* configuration with two *DRPs*.

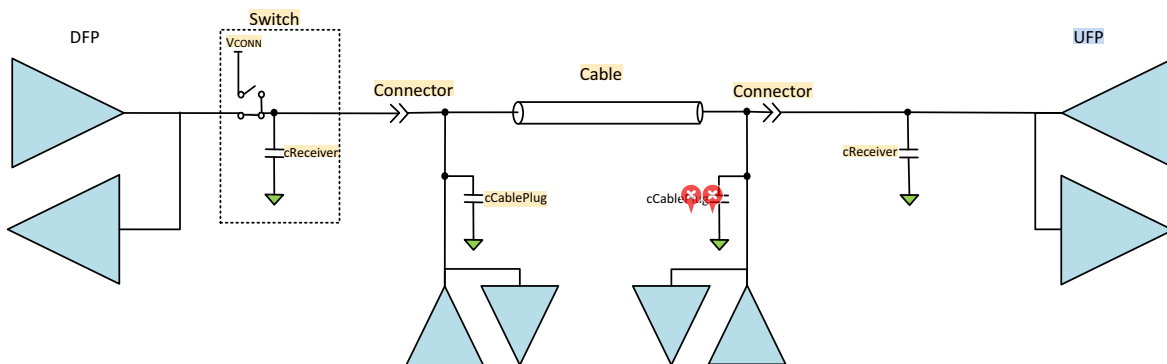
Figure 5.27 Example Multi-Drop Configuration showing two DRPs



The *Multi-Drop* transceiver **shall** obey all the electrical characteristics specified in this section except for those relating to capacitance. The maximum capacitance allowed for the *Multi-Drop* node when not driving the line is *cCablePlug_{CC}* defined in [USB Type-C 2.4]. There are no constraints as to the distance of the *Multi-Drop* transceiver from the end of the plug. The *Multi-Drop* transceiver(s) **may** be located anywhere along the cable including the plugs. The *Multi-Drop* transceiver suffers less from ground offset compared to the transceivers in the *USB Host or USB Device* and contributes no significant reflections.

It is possible to have a configuration at *Attach* where one *Port* can be a *VCONN Source* and the other *Port* is not able to be a *VCONN Source*, such that there is no switch in the second *Port*. An example of a *DFP* with a switch *Attached* to a *UFP* without a switch is outlined in Figure 5.28, "Example Multi-Drop Configuration showing a DFP and UFP". The capacitance on the *CC* line for a *Port* not able to be a *VCONN Source* **shall** still be within *cReceiver* except when transmitting.

Figure 5.28 Example Multi-Drop Configuration showing a DFP and UFP



5.8.6.3 Fast Role Swap Detection

An *Initial Sink* prepares for a *Fast Role Swap* by ensuring that once it has detected the *Fast Role Swap Request* its power supply is ready to respond by applying *vSafe5V* according to the timing detailed in [Section 7.1.13, "Fast Role Swap"](#). The *Initial Sink* **shall** only respond to the *Fast Role Swap Request* when all the following conditions have been met:

- An *Explicit Contract* has been established and the *Sink Capabilities* of the *Initial Source* have been received by, and at the request of, the *Initial Sink*.
- The *Sink Capabilities* Message received from the *Initial Source* has at least one of the *Fast Role Swap* bits set in its 5V *Fixed Supply* PDO.
- The *Initial Sink* is able and willing to source the current requested by the *Initial Source* in the *Fast Role Swap* bits of its *Sink Capabilities* Message.

On detection of the *Fast Role Swap Request* any pending Messages **shall** be **Discarded** (see [Section 6.12.2.2.1, "Common Protocol Layer Message Transmission State Diagram"](#)).

When the *Initial Sink* is prepared for a *Fast Role Swap* and the bus is idle the *CC* voltage averaged over *tFRSwapRx* min remains above 0.7V (see [\[USB Type-C 2.4\]](#)) since the *Source* R_p is either 1.5A or 3.0A. However, *vNoiseIdle* noise **may** cause the *CC* line voltage to reach $0.7V - vNoiseIdle/2$ for short durations. When the *Initial Sink* is prepared for a *Fast Role Swap* while it is transmitting and the *Initial Source* is sending a *Fast Role Swap Request*, the transmission will be attenuated such that the peak *CC* voltage will not exceed *vFRSwapCableTx* min. Therefore, when the *Initial Sink* is prepared for a *Fast Role Swap*, it **shall not** detect a *Fast Role Swap Request* when the *CC* voltage, averaged over *tFRSwapRx* min, is above 0.7V. When the *Initial Sink* is prepared for a *Fast Role Swap*, it **shall** detect a *CC* voltage lower than *vFRSwapCableTx* min for *tFRSwapRx* as a *Fast Role Swap Request*.

Note: The *Initial Sink* is not required to average the *CC* voltage to meet these requirements.

The *Initial Sink* **shall** initiate the *Fast Role Swap* AMS within *tFRSwapInit* of detecting the *Fast Role Swap Request* in order to assign the R_p/R_d resistors to the correct *Ports* and to re-synchronize the state machines (see [Section 6.3.19, "FR Swap Message"](#)).

The *Initial Sink* **shall** become the *New Source* and **shall** start supplying *vSafe5V* at *USB Type-C* current (see [\[USB Type-C 2.4\]](#)) no later than *tSrcFRSwap* after *VBUS* has dropped below *vSafe5V*. An *Initial Sink* **shall** disable its *VBUS* Disconnect Threshold detection circuitry while *Fast Role Swap* detection is active.

Note: While power is transitioning the *VCONN Source* to the *Cable Plug(s)* cannot be guaranteed.

5.9 Built in Self-Test (BIST)

The following sections define *BIST* functionality which **Shall** be supported.

5.9.1 BIST Carrier Mode

In *BIST Carrier Mode*, the *PHY Layer* **Shall** send out a *BMC* encoded continuous string of alternating "1"s and "0"s. This enables the measurement of power supply noise and frequency drift.

Note: This transmission is a purely a sequence of alternating bits and **Shall Not** be formatted as a *Packet*.

See also [Section 6.4.3, "BIST Message"](#).

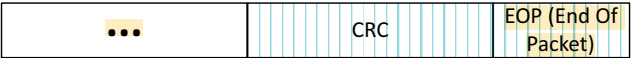
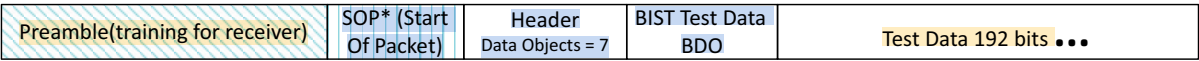


5.9.2 BIST Test Data Mode

A *BIST Test Data* Message is used by the *Tester* to send various *Tester* generated *Test Patterns* to the *UUT* in order to test the *UUT*'s receiver. See also [Section 6.4.3, "BIST Message"](#).

[Figure 5.29, "Test Frame"](#) shows the *Test Frame* which **Shall** be sent by the *Tester* to the *UUT*. The *BIST* Message, with a *BIST Test Data* BIST Data Object consists of a *Preamble*, followed by *SOP**, followed by the *Message Header* with a data length of 7 *Data Objects*, followed a *BIST Test Data* BIST Data Object, followed by 6 *Data Objects* containing *test* data, followed by the *CRC* and then an *EOP*.

Figure 5.29 Test Frame



LEGEND:

